

Ankit Kumar

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Full-Stack AI & Backend Engineer specializing in Java, LLM-integrated applications, and production-grade REST APIs. Built a stateless Medical RAG pipeline and real-time backend systems; focused on architecture, zero-defect data flows, and AI integration.

TECHNICAL SKILLS

Languages & Frameworks	Java (Primary), Python, Spring Boot, FastAPI, JavaScript, React, SQL
AI & LLM	LangChain, RAG Pipelines, Prompt Engineering, Gemini, Vector DBs (ChromaDB)
Backend & Systems	REST APIs, WebSockets, WebRTC
Database & Tools	PostgreSQL, Git

ACHIEVEMENTS

- **TCS CodeVita Season 13 — Global Rank 15** out of 669,000+ participants
- **Codeforces Expert** (rating 1655) · **LeetCode** — active problem solver across DSA domains

WORK EXPERIENCE

Software Engineer Intern

Feb 2025 – May 2025

ControlShift Talent Pvt. Ltd.

- Architected the backend workflow for an ATS platform, integrating resume filtering, admin scheduling, and live AI interviewing sessions using WebRTC and WebSockets for real-time candidate communication
- Collaborated with the internal AI team to implement automated candidate cheat detection; built a system to chunk live video recordings, queue the data locally, and seamlessly pass it to the asynchronous AI processing pipeline

PROJECTS

Arogya AI — Stateless Medical RAG Engine

github.com/ankitkr-04/arogya_ai

Python, FastAPI, LangChain, ChromaDB, Gemini

- Engineered an agentic, stateless REST API workflow utilizing FastAPI and LangChain to process medical queries, generating dynamic follow-up questions when the initial context lacked sufficient symptom detail
- Integrated ChromaDB for Retrieval-Augmented Generation (RAG), dynamically routing queries between optimistic (general advice) and pessimistic (diagnostic consideration) vector collections based on automated severity assessment
- Implemented strict LLM guardrails via multi-stage prompt engineering and Pydantic validation to flag clinical red-flags, prevent diagnostic hallucinations, and enforce structured JSON outputs

TicketLedger — High-Concurrency Cinema Booking Platform

github.com/ankitkr-04/ticket-ledger

Java 21, Spring Boot, PostgreSQL, Stripe API

- Enforced zero double-bookings via PostgreSQL `SELECT FOR UPDATE` with strict lock ordering; analytically bounded hot-seat throughput at **~200 ops/sec** per row from lock-commit latency
- Scaled API layer by isolating database connection pools (HikariCP) to handle large volumes of parked connections per JVM without exhausting OS threads
- Guaranteed exactly-once payment execution via idempotent Stripe webhook processing, prioritizing strict financial consistency over UX in asynchronous late-payment edge cases

Axiom — Redis-Compatible In-Memory Store

github.com/ankitkr-04/axiom

Java, NIO, Concurrency

- Delivered **~81K GET ops/sec** via a single-threaded Java NIO event loop with direct `ByteBuffer` access, eliminating thread contention and Garbage Collection pressure
- Implemented 29+ Redis-compatible commands including Sorted Sets (skip-list) and Geospatial (geohash)
- Replicated Redis PSYNC protocol: executed RDB snapshots on initial sync with async incremental streaming

EDUCATION

B.Tech in Computer Science

Aug 2023 – May 2026

Lakshmi Narain College of Technology, Bhopal

CGPA: **8.38 / 10**

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Developer Professional (1Z0-1084-25) — Oracle